Series of Real Numbers

Convergence, Divergence, and Properties

What is a Series?

A series is the sum of the terms of a sequence. Given a sequence $\{a_n\}$, the series is denoted by:

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$$

The partial sum S_N of the series is the sum of the first N terms:

$$S_N = \sum_{n=1}^{N} a_n = a_1 + a_2 + \cdots + a_N$$

A series converges if the sequence of partial sums $\{S_N\}$ converges to a finite limit S: $\lim_{N\to\infty} S_N = S$.
If the limit does not exist or is infinite, the series diverges.

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Examples of Convergence and Divergence

The geometric series $\sum_{n=0}^{\infty} ar^n$ converges to $\dfrac{a}{1-r}$ if $|r| < 1$ and diverges if $|r| \geq 1$.

The harmonic series $\sum_{n=1}^{\infty} \dfrac{1}{n}$ diverges, even though $\lim_{n\to\infty} \dfrac{1}{n} = 0$.

The series $\sum_{n=1}^{\infty} \left(\dfrac{1}{n} - \dfrac{1}{n+1}\right)$ converges to 1:

$$S_N = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{N} - \frac{1}{N+1}\right) = 1 - \frac{1}{N+1}$$

$$\lim_{N\to\infty} S_N = 1$$

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Necessary Condition for Convergence

Theorem

If the series $\sum_{n=1}^{\infty} a_n$ *converges, then* $\lim_{n\to\infty} a_n = 0$.

Proof

Let $S = \lim_{N\to\infty} S_N$. Then

$$a_n = S_n - S_{n-1} \quad \Rightarrow \quad \lim_{n\to\infty} a_n = \lim_{n\to\infty} (S_n - S_{n-1}) = S - S = 0.$$

Important Note

The converse is not true. If $\lim_{n\to\infty} a_n = 0$, the series may still diverge (e.g., the harmonic series).

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Counterexample: Harmonic Series

[Harmonic Series - Divergence Proof] The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ satisfies $\lim_{n\to\infty} \frac{1}{n} = 0$ but diverges.

Group the terms:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots$$

Each group is greater than $\frac{1}{2}$:

$$\frac{1}{3} + \frac{1}{4} > \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > 4 \times \frac{1}{8} = \frac{1}{2}$$

So the partial sums grow without bound, proving divergence.

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Addition of Series

Theorem

If $\sum a_n$ and $\sum b_n$ are convergent series with sums A and B respectively, then:

1. $\sum(a_n + b_n)$ *converges to* $A + B$
2. $\sum ca_n$ *converges to* cA *for any constant* c

$$\sum_{n=1}^{\infty}\left(\frac{1}{2^n} + \frac{1}{n(n+1)}\right) = \sum_{n=1}^{\infty}\frac{1}{2^n} + \sum_{n=1}^{\infty}\frac{1}{n(n+1)} = 1 + 1 = 2$$

Warning

If both series diverge, their sum may converge or diverge. If one converges and one diverges, their sum always diverges.

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Series of Positive Terms

A series $\sum a_n$ is called a series of positive terms if $a_n > 0$ for all n.

For a series of positive terms, the sequence of partial sums $\{S_N\}$ is increasing.

An increasing sequence converges if and only if it is bounded above.

A series of positive terms converges if and only if its partial sums are bounded above.

The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges because the partial sums are bounded above (by 2, in fact).

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Absolute and Conditional Convergence

A series $\sum a_n$ **converges absolutely** if $\sum |a_n|$ converges.

A series that converges but does not converge absolutely is said to **converge conditionally**.

If a series converges absolutely, then it converges.

The alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges conditionally to $\ln 2$.

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Rearrangements of Terms

A rearrangement of a series $\sum a_n$ is a series $\sum a_{\sigma(n)}$ where σ is a permutation of $\mathbb{N}$.

If a series converges absolutely, then all its rearrangements converge to the same sum.

If a series converges conditionally, then for any real number L, there exists a rearrangement that converges to L.

The alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges conditionally to $\ln 2$, but can be rearranged to converge to any real number.

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Comparison Test (First Type)

Theorem 6.3.1 Let $\sum u_n$ and $\sum v_n$ be two series of positive terms. If there exists a constant $k > 0$ such that

$$u_n \leq kv_n \quad \text{for all large } n,$$

then

$$\begin{cases} \text{If } \sum v_n \text{ converges, then } \sum u_n \text{ converges.} \\ \text{If } \sum u_n \text{ diverges, then } \sum v_n \text{ diverges.} \end{cases}$$

Example: Comparison Test

Test the convergence of

$$\sum_{n=1}^{\infty} \frac{1^2 + 2^2 + \cdots + n^2}{n^4 + n}.$$

Solution: We have

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

So

$$u_n = \frac{\frac{n(n+1)(2n+1)}{6}}{n^4 + n} \sim \frac{1}{3n}.$$

Take $v_n = \frac{1}{n}$. Since $\sum \frac{1}{n}$ diverges, $\sum u_n$ also diverges.

Limit Comparison Test

Theorem 6.3.2 Let $\sum u_n$ and $\sum v_n$ be two series of positive terms. If

$$\lim_{n\to\infty} \frac{u_n}{v_n} = l, \quad 0 < l < \infty,$$

then both $\sum u_n$ and $\sum v_n$ converge or diverge together.

Example: Limit Comparison Test

Test the convergence of

$$\sum_{n=1}^{\infty} \frac{n}{n^3 + 1}.$$

Solution: Take $v_n = \frac{1}{n^2}$. Then

$$\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{\frac{n}{n^3+1}}{\frac{1}{n^2}} = \lim_{n\to\infty} \frac{n^3}{n^3 + 1} = 1.$$

Since $\sum \frac{1}{n^2}$ converges, $\sum \frac{n}{n^3+1}$ also converges.

Ratio Test

Theorem 6.3.4 Let $\sum u_n$ be a series of positive terms. If

$$\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = l,$$

then

$$\begin{cases} \sum u_n \text{ converges if } l < 1, \\ \sum u_n \text{ diverges if } l > 1. \end{cases}$$

If $l = 1$, the test is inconclusive.

Example: Ratio Test

Test the convergence of

$$\sum_{n=1}^{\infty} \frac{n^2}{n!}.$$

Solution:

$$\frac{u_{n+1}}{u_n} = \frac{(n+1)^2}{(n+1)!} \cdot \frac{n!}{n^2} = \frac{(n+1)^2}{(n+1)n^2} \to 0.$$

So the series converges by ratio test.

Root Test

Theorem 6.3.5 Let $\sum u_n$ be a series of positive terms. If

$$\lim_{n\to\infty} \sqrt[n]{u_n} = l,$$

then

$$\sum u_n \text{ converges if } l < 1, \quad \text{diverges if } l > 1.$$

If $l = 1$, the test is inconclusive.

Example: Root Test

Test the convergence of

$$\sum_{n=1}^{\infty}\left(\frac{2n}{2n+1}\right)^{n}.$$

Solution:

$$\sqrt[n]{u_n} = \frac{2n}{2n+1} \to 1.$$

Here the root test fails (inconclusive). Another test is needed.

Test the convergence of

$$\sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n .$$

Solution: Let

$$u_n = \left(\frac{3}{4}\right)^n .$$

Then

$$\sqrt[n]{u_n} = \sqrt[n]{\left(\frac{3}{4}\right)^n} = \frac{3}{4}.$$

So

$$\lim_{n\to\infty} \sqrt[n]{u_n} = \tfrac{3}{4} < 1.$$

Conclusion: The series converges absolutely by the Root Test.

Theorem 6.3.6 Let $\sum u_n$ be a series of positive terms and suppose

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = l.$$

Then

$$\begin{cases} \sum u_n \text{ converges if } l > 1, \\ \sum u_n \text{ diverges if } l < 1. \end{cases}$$

If $l = 1$, the test is inconclusive.

Theorem 6.4.1 (Leibniz Test) If $\{u_n\}$ is a decreasing sequence of positive numbers with $\lim_{n\to\infty} u_n = 0$, then the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} u_n$$

converges.

Example: Leibniz Test

Test the convergence of

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}.$$

Solution: Here $u_n = \frac{1}{n}$ which decreases to 0.
Therefore, by Leibniz test, the alternating harmonic series converges.

Abel's Test

Theorem 6.4.2 (Abel's Test) If $\{a_n\}$ is a bounded monotonic sequence and $\sum b_n$ is a convergent series of real numbers, then

$$\sum a_n b_n$$

converges.

Dirichlet's Test

Theorem 6.4.3 (Dirichlet's Test) If $\{a_n\}$ is a monotonic sequence tending to 0, and the partial sums of $\sum b_n$ are bounded, then

$$\sum a_n b_n$$

converges.

Example: Dirichlet's Test

Test the convergence of

$$\sum_{n=1}^{\infty} \frac{\sin n}{n}.$$

Solution:

- Take $a_n = \frac{1}{n}$ and $b_n = \sin n$.
- a_n is a positive decreasing sequence with $\lim a_n = 0$.
- The partial sums of $\sum \sin n$ are bounded (well-known result).

Conclusion

By Dirichlet's Test,

$$\sum_{n=1}^{\infty} \frac{\sin n}{n}$$

converges.

The p-Series

Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots$$

Result

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \begin{cases} \text{converges,} & p > 1, \\ \text{diverges,} & 0 < p \leq 1. \end{cases}$$

Examples:

- $p = 1$: Harmonic series diverges.
- $p = 2$: Converges to $\frac{\pi^2}{6}$.

Definition: Let Σu_n be a series of positive and negative real numbers. Let $u_n' = |u_n|$. Then $\Sigma u_n'$ is a series of positive real numbers. If $\Sigma u_n'$ is convergent then Σu_n is said to be an **absolutely convergent series**.

Theorem: An absolutely convergent series is convergent.
Proof: Let Σu_n be absolutely convergent. Then $\Sigma|u_n|$ converges.
By Cauchy's principle, for any $\epsilon > 0$, $\exists m$ such that:

$$|u_{n+1}| + |u_{n+2}| + \cdots + |u_{n+p}| < \epsilon$$

for all $n \geq m$ and every natural number p.
Since:

$$|u_{n+1} + u_{n+2} + \cdots + u_{n+p}| \leq |u_{n+1}| + |u_{n+2}| + \cdots + |u_{n+p}| < \epsilon$$

Therefore, Σu_n converges.

Examples of Absolute Convergence

1. The series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots$ is absolutely convergent.
2. For fixed x, the series $\sum \frac{\sin nx}{n}$ is absolutely convergent since:

$$\left| \frac{\sin nx}{n} \right| \leq \frac{1}{n}$$

and $\sum \frac{1}{n}$ converges.

Theorem 6.4.2

Theorem: If Σu_n is absolutely convergent and $\{v_n\}$ is bounded, then $\Sigma u_n v_n$ is absolutely convergent.
Proof: There exists $k > 0$ such that $|v_n| < k$ for all n.
Then:

$$|u_{n+1} v_{n+1}| + \cdots + |u_{n+p} v_{n+p}| < k(|u_{n+1}| + \cdots + |u_{n+p}|)$$

Since $\Sigma |u_n|$ converges, by Cauchy's principle, the right side can be made arbitrarily small. Hence, $\Sigma |u_n v_n|$ converges.

Worked Example 1

Test the series: $\frac{2}{3} - \frac{3}{4} + \frac{4}{5} - \frac{5}{6} + \cdots$

Solution: Let $u_n = (-1)^{n+1} \frac{n+1}{n^2} = \frac{(-1)^{n+1}}{n^2} \left(1 + \frac{1}{n}\right) = a_n b_n$

where $a_n = \frac{(-1)^{n+1}}{n^2}$, $b_n = 1 + \frac{1}{n}$.

Σa_n is absolutely convergent, $\{b_n\}$ is bounded. Hence, Σu_n is absolutely convergent.

Theorem 6.4.3 (Ratio Test)

Theorem: Let Σu_n be a series with $\lim \frac{|u_{n+1}|}{|u_n|} = l$.

- Absolutely convergent if $l < 1$
- Divergent if $l > 1$

Proof of Ratio Test

(i) $l < 1$: Choose $\epsilon > 0$ such that $l + \epsilon = r < 1$.
Then $\exists m$ such that:

$$\frac{|u_{n+1}|}{|u_n|} < r \quad \forall n \geq m$$

Hence:

$$|u_n| < |u_m| r^{n-m}$$

Compare with geometric series Σr^n (convergent).
(ii) $l > 1$: Then $\frac{|u_{n+1}|}{|u_n|} > 1$ for large n, so $\{|u_n|\}$ increases and $\lim u_n \neq 0$.
Hence, divergent.

Theorem 6.4.4 (Root Test)

Theorem: Let Σu_n be a series with $\lim |u_n|^{1/n} = l$.

- Absolutely convergent if $l < 1$
- Divergent if $l > 1$

Proof of Root Test

(i) $l < 1$: Choose $\epsilon > 0$ such that $l + \epsilon = r < 1$.
Then $\exists m$ such that:

$$|u_n|^{1/n} < r \quad \forall n \geq m$$

Hence:

$$|u_n| < r^n$$

Compare with geometric series Σr^n (convergent).
(ii) $l > 1$: Then $|u_n| > 1$ for large n, so $\lim u_n \neq 0$. Hence, divergent.

Worked Example 2

Examine convergence of: $1 - \frac{a^2}{2!} + \frac{a^3}{3!} - \frac{a^4}{4!} + \cdots$

Solution: $u_n = (-1)^{n+1} \frac{a^n}{n!}$

$$\frac{|u_{n+1}|}{|u_n|} = \frac{a}{n+1} \to 0 < 1$$

Hence, absolutely convergent.

Worked Example 3

Examine convergence of: $1 - \frac{(2!)^2}{4!} + \frac{(3!)^2}{6!} - \cdots$

Solution: $u_n = (-1)^{n+1} \frac{(n!)^2}{(2n)!}$

$$\frac{|u_{n+1}|}{|u_n|} = \frac{(n+1)^2}{(2n+1)(2n+2)} \to \frac{1}{4} < 1$$

Hence, absolutely convergent.

Definition: If $u_n > 0$ for all n, the series $\Sigma(-1)^{n+1}u_n$ is called an **alternating series**.

Theorem 6.4.5 (Leibniz's Test)

Theorem: If $\{u_n\}$ is monotone decreasing and $\lim u_n = 0$, then the alternating series:

$$u_1 - u_2 + u_3 - u_4 + \cdots$$

converges.

Proof of Leibniz's Test

Let $s_n = u_1 - u_2 + u_3 - \cdots + (-1)^{n+1} u_n$.

- $s_{2n+2} - s_{2n} = u_{2n+1} - u_{2n+2} \geq 0$: $\{s_{2n}\}$ increasing
- $s_{2n+1} - s_{2n-1} = -u_{2n} + u_{2n+1} \leq 0$: $\{s_{2n+1}\}$ decreasing
- $s_{2n} < u_1$: bounded above
- $s_{2n+1} > u_1 - u_2$: bounded below

Both subsequences converge to the same limit since:

$$\lim(s_{2n+1} - s_{2n}) = \lim u_{2n+1} = 0$$

Hence, $\{s_n\}$ converges.

Error Estimate

If s is the sum and s_n the nth partial sum, then:

$$0 < (-1)^n(s - s_n) < u_{n+1}$$

Examples of Leibniz's Test

1. $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ converges.
2. $\frac{1}{1+a} - \frac{1}{2+a} + \frac{1}{3+a} - \cdots$ converges for $a > -1$.

Theorem 6.4.6 (Abel's Test)

Theorem: If $\{b_n\}$ is monotone bounded and Σa_n converges, then $\Sigma a_n b_n$ converges.

Proof of Abel's Test

Let $s_n = a_1 + \cdots + a_n$, $t_n = a_1 b_1 + \cdots + a_n b_n$.
Then:

$$t_n = s_1(b_1 - b_2) + s_2(b_2 - b_3) + \cdots + s_n(b_n - b_{n+1}) + s_n b_{n+1}$$

- $s_n b_{n+1}$ converges (since s_n convergent, b_n bounded)
- $\Sigma s_n(b_n - b_{n+1})$ absolutely convergent (since $\Sigma|b_n - b_{n+1}|$ convergent)

Hence, t_n converges.

Theorem 6.4.7 (Dirichlet's Test)

Theorem: If $\{b_n\}$ is monotone with $\lim b_n = 0$ and the partial sums of Σa_n are bounded, then $\Sigma a_n b_n$ converges.

Proof of Dirichlet's Test

Similar to Abel's test:

$$t_n = s_1(b_1 - b_2) + \cdots + s_n(b_n - b_{n+1}) + s_n b_{n+1}$$

- $s_n b_{n+1} \to 0$
- $\Sigma s_n(b_n - b_{n+1})$ absolutely convergent

Hence, t_n converges.

Note

Leibniz's test is a special case of Dirichlet's test with:

- $a_n = (-1)^{n+1}$ (bounded partial sums)
- $b_n = u_n$ (monotone decreasing to 0)

Examples

1. $\sum \frac{(-1)^{n+1}}{\sqrt{n}}$ converges by Dirichlet's test.
2. $\sum \frac{(-1)^{n+1}}{\log(n+1)}$ converges by Dirichlet's test.